

Chikungunya transmission and population susceptibility in Ceará, Brazil (2015-2025)

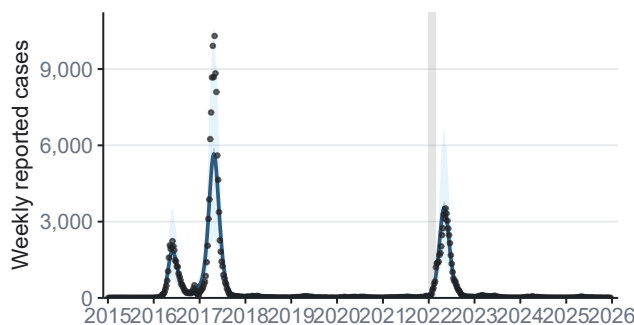
v4.9 hierarchical-seasonality sweep, $q = 0.05$ (HMC FAIL/marginal) | 2022 seed conditioned, year-varying seasonal amplitude

A

Reported cases and posterior prediction

Grey band = 2022 seed window (conditioned, exclude)

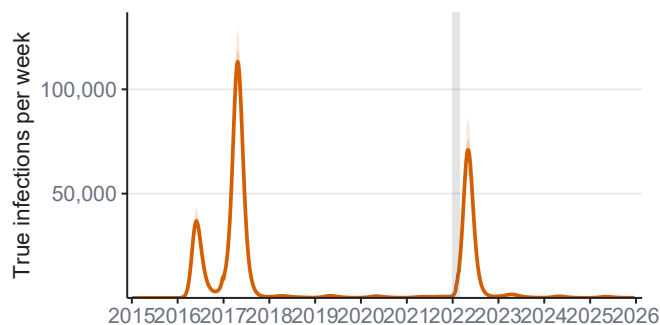
— Model expectation • Observed cases



B

Latent infection incidence

Grey band = 2022 seed window ($X = C/q$, conditioned)



C

Population susceptibility and immunity

Closed population; LIFELONG immunity; continuous threshold

— Infection-derived immune — Susceptible

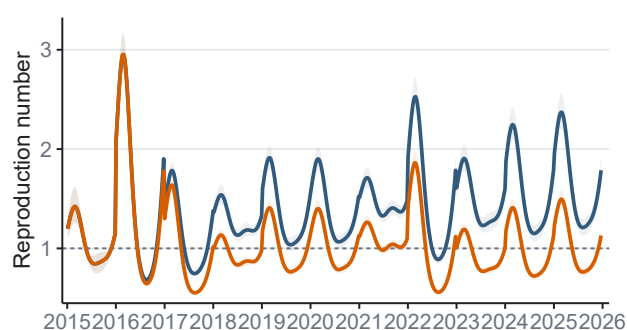


D

Time-varying reproduction numbers

Hierarchical seasonal amplitude: A_{year} modulates the first

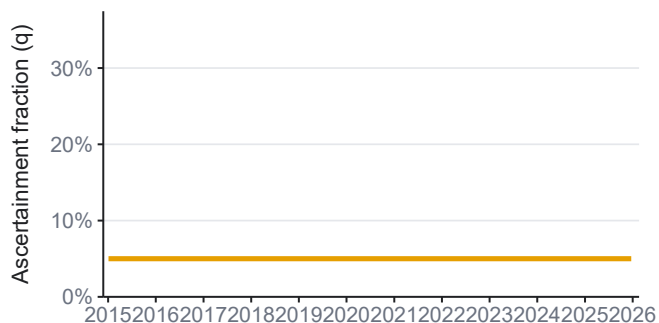
— $R_0(t)$ — $R_{\text{eff}}(t)$



E

Case ascertainment

q FIXED at 0.05 (not estimated -- v4.9 hierarchical-seas)



F

Cumulative infection and serological anchor

Yellow field: June-December 2018 survey

— Ceara immune fraction — Model Ceara immune fraction
— Observed Juazeiro seroprevalence

